CAFETERIA MANAGEMENT MODULE

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 · March 2026
Confidential · Internal Documentation

Audience: Development Team · Product Team · School Administrators
Related Modules: Student Management · Finance · Inventory · Parent Portal · Staff Management · Attendance

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1. MODULE OVERVIEW

1.1 Purpose

The **Cafeteria Management Module** manages all school food service operations including:

- Student meal plans
- Daily meal distribution
- Menu management
- Cafeteria payments
- Dietary management
- Kitchen inventory
- Nutrition tracking
- Parent/student meal interactions

The module ensures **efficient meal distribution, transparency, financial accountability, and health compliance**.

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2. CORE OBJECTIVES

The system must enable schools to:

1. Manage **daily meals and menus**
2. Track **student meal participation**
3. Handle **meal plan subscriptions**
4. Support **dietary restrictions and allergies**
5. Integrate **payments and billing**
6. Monitor **kitchen inventory**
7. Allow **student/parent interaction**

8. Provide **analytics for administrators**

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3. SYSTEM ROLES & ACCESS CONTROL

Role	Permissions
Super Admin	Full configuration and oversight
School Admin	Manage cafeteria operations
Cafeteria Staff	Manage meals, logs, and distribution
Finance Staff	Handle billing and payments
Students	View menu and meal status
Parents	Manage meal plans and payments
Teachers	Monitor student meal participation

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4. MODULE ARCHITECTURE

The Cafeteria Module is divided into the following subsystems:

```
Cafeteria Management
│
├── Cafeteria Dashboard
├── Menu Management
├── Meal Plans
├── Daily Meal Logs
├── Student Meal Accounts
├── Dietary & Allergy Management
├── Pre-Orders
├── Cafeteria Payments
├── Kitchen Inventory
├── Cafeteria Reports & Analytics
```

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5. CAFETERIA DASHBOARD

5.1 Overview

The **Cafeteria Dashboard** provides a real-time overview of daily food operations.

Dashboard Widgets

- Today's Menu

- Meals Prepared
- Meals Served
- Students on Meal Plan
- Inventory Alerts
- Revenue Today
- Dietary Alerts

Example UI Layout

```
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Cafeteria Dashboard
-----------------------------------------------

Today's Menu
Breakfast: Pancakes, Milk
Lunch: Rice, Chicken Stew, Vegetables
Snack: Fruit

Meals Prepared: 520
Meals Served: 468

Students with Active Meal Plan: 700

Inventory Alerts:
 Milk running low
 Bread stock low

Revenue Today: $1,240
```

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6. MENU MANAGEMENT

6.1 Purpose

Allows cafeteria administrators to define **daily and weekly menus**.

6.2 Features

- Create daily meals
- Define weekly menu schedule
- Attach nutritional information
- Define meal pricing
- Mark meals for dietary types

6.3 Menu Data Model

```
Menu
```

```
- menu_id
- school_id
- date
- meal_type (breakfast/lunch/snack)
- meal_name
- ingredients
- calories
- allergens
- price
```

6.4 Weekly Menu Planner

Admins can pre-plan meals.

Example:

```
Monday
Breakfast: Oatmeal
Lunch: Rice + Beans
Snack: Apple

Tuesday
Breakfast: Pancakes
Lunch: Chicken + Vegetables
Snack: Yogurt
```

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7. MEAL PLANS

7.1 Overview

Students can subscribe to **meal plans**.

Types:

- Daily plan
- Weekly plan
- Monthly plan
- Pay-as-you-eat

7.2 Plan Structure

```
MealPlan
- plan_id
- plan_name
```

```
- duration
- meals_included
- price
```

7.3 Student Enrollment

```
StudentMealPlan
- student_id
- plan_id
- start_date
- end_date
- status
```

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8. DAILY MEAL LOGS

8.1 Purpose

Track meals served each day.

8.2 Data Model

```
MealLog
- log_id
- date
- student_id
- meal_type
- meal_id
- served_by
- time_served
```

8.3 Meal Distribution Methods

The system may support:

1. **Student ID scanning**
2. **QR code scanning**
3. **Manual attendance**
4. **Biometric check-in**

Example:

```
Student scans ID card → system verifies meal plan → meal served logged
```

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9. STUDENT MEAL ACCOUNTS

Each student has a **cafeteria wallet/account**.

Features

- Balance tracking
- Meal history
- Recharge options
- Payment alerts

Data Model

```
MealAccount
- student_id
- balance
- total_spent
- last_transaction
```

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10. DIETARY & ALLERGY MANAGEMENT

Critical for school health compliance.

10.1 Supported Dietary Types

- Vegetarian
- Vegan
- Halal
- Gluten-Free
- Lactose-Free

10.2 Allergy Tracking

Example:

```
Student Allergies
- Nuts
- Dairy
- Eggs
```

When a restricted food appears on the menu:

```
System Alert:
 Student allergic to nuts
Alternative meal recommended
```

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11. MEAL PRE-ORDER SYSTEM

Students or parents can **pre-order meals**.

Benefits

- Reduces food waste
- Improves kitchen planning

11.1 Workflow

```
Parent opens Parent Portal
→ views tomorrow's menu
→ selects meal
→ confirms order
```

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12. CAFETERIA PAYMENTS

12.1 Payment Methods

- Cash
- School wallet
- Online payment
- Mobile money
- Bank transfer

12.2 Finance Integration

Connected to **Finance Module**

```
Finance Module
    ↓
Cafeteria Billing
    ↓
Student Account
```

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13. KITCHEN INVENTORY MANAGEMENT

Connected to **Store & Inventory Module**.

Tracks:

- Ingredients
- Supplies
- Stock levels

13.1 Inventory Data Model

```
Ingredient
- ingredient_id
- name
- quantity
- unit
- supplier
- expiry_date
```

13.2 Automatic Stock Deduction

When meals are served:

```
Chicken Stew
→ chicken -2kg
→ vegetables -1kg
→ oil -0.5L
```

Inventory updates automatically.

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14. PARENT & STUDENT PORTAL FEATURES

Parent Portal

Parents can:

- View weekly menu
- Manage meal plans
- Top-up cafeteria wallet
- See meal history
- Set dietary restrictions
- Pre-order meals

Student Portal

Students can:

- View today's menu
- Check meal balance
- View meal history
- Pre-order meals

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15. REPORTS & ANALYTICS

Cafeteria Reports

Admins can generate:

Financial Reports

- Daily revenue
- Meal plan subscriptions
- Payment summaries

Operational Reports

- Meals served
- Food consumption
- Waste analysis

Student Reports

- Meal participation
- Dietary alerts

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16. NOTIFICATION SYSTEM

Integrated with the **Communication Module**.

Notifications:

- Low meal balance
- Menu updates
- Payment reminders
- Dietary alerts

Channels:

- In-app
- Email
- SMS
- Push notifications

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17. SECURITY & DATA PROTECTION

Access Control

Role-based access:

```
Admin → full control
Staff → meal distribution
Parents → child meals
Students → personal data
```

Data Protection

- Encrypted financial data
- Secure authentication
- Audit logs

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18. SYSTEM INTEGRATIONS

Module	Integration
Student Management	Student records
Finance	Payments
Inventory	Ingredients
Communication	Notifications
Parent Portal	Meal management
Attendance	Meal tracking

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19. FUTURE ENHANCEMENTS

Recommended upgrades:

- AI-based meal demand prediction
- Nutrition analytics
- Food waste tracking
- Smart cafeteria cards
- Facial recognition meal check-in
- Mobile ordering app